

Software Requirements – Shop Management System

1. Introduction

The Shop Management System is a software system used to manage the daily activities of a shop. It stores customer and product information, records sales, applies discounts, and generates the final bill.

2. Objective of the System

The main objectives are:

To maintain customer details.

To store and manage product details.

To record product sales.

To calculate and apply discounts.

To calculate the final bill accurately.

To reduce manual work and calculation errors.

3. Functional Requirements

3.1 Customer Details

The system should allow the user to enter customer ID, name, phone number, and address.

The system should store customer information.

The user should be able to view and update customer details.

3.2 Product Details

The system should store product ID, product name, price, and stock.

The user should be able to view product details.

The system should check product availability before a sale.

3.3 Sale

The user should select the required product.

The user should enter the quantity.

The system should calculate the sale amount.

Sale Amount = Price × Quantity

3.4 Discount

The system should allow the user to enter or apply a discount percentage.

The system should calculate the discount amount.

Discount Amount = Sale Amount × Discount % / 100

3.5 Final Bill

The system should calculate the final amount after discount.

Final Bill = Sale Amount – Discount Amount

The system should display customer details, product details, quantity, discount, and final amount.

The system should generate the final bill.

4. Non-Functional Requirements

The system should be easy to use.

Calculations should be accurate.

The system should provide quick responses.

Data should be stored securely.

The system should be reliable.

The interface should be simple and user-friendly.

5. Hardware Requirements

Computer or laptop

Minimum 4 GB RAM

Minimum 20 GB free storage

Keyboard and mouse

Monitor/display

Printer (optional for printing bills)

6. Software Requirements

Operating System: Windows / Linux

Programming Language: C / C++ / Java / Python

Database: MySQL / SQLite

Code editor or IDE

Required database connectivity software

7. User Requirements

The user should be able to:

Add and manage customer details.

Add and manage product details.

Check product stock.

Record sales.

Apply discounts.

Calculate the final bill.

View and print/save the final bill.

8. System Constraints

Correct customer and product information must be entered.

A product must be available in stock before making a sale.

The system requires suitable hardware and software.

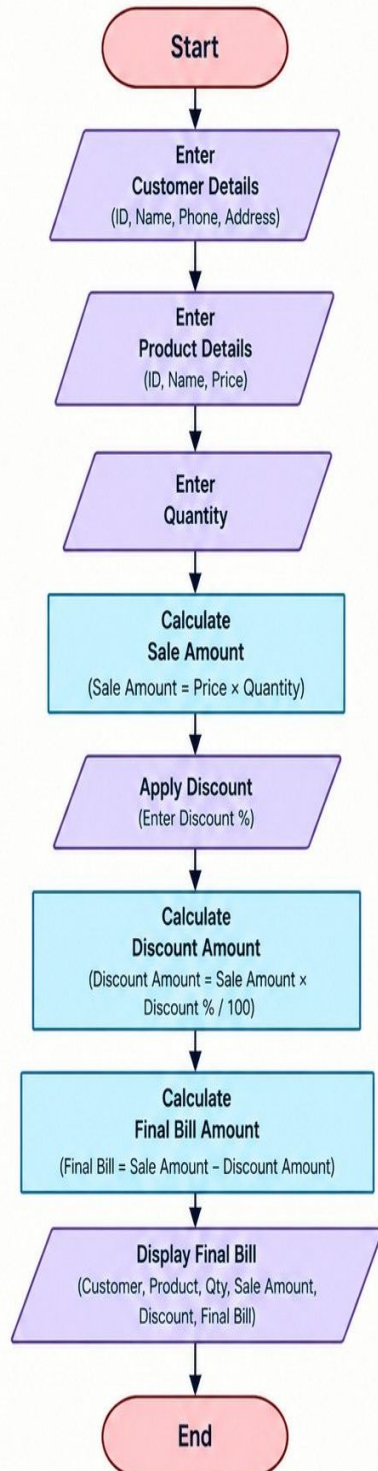
Only authorized users should modify important records.

The system depends on the selected database and programming environment.

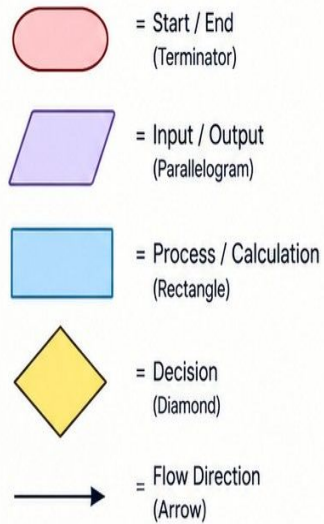
Algorithm – Shop Management System

1. Start
2. Enter customer details such as Customer ID, Name, Phone Number, and Address.
3. Enter product details such as Product ID, Product Name, and Price.
4. Enter the quantity of the product.
5. Calculate the Sale Amount:
 $\text{Sale Amount} = \text{Price} \times \text{Quantity}$
6. Enter or apply the Discount Percentage.
7. Calculate the Discount Amount:
 $\text{Discount Amount} = \text{Sale Amount} \times \text{Discount Percentage} \div 100$
8. Calculate the Final Bill Amount:
 $\text{Final Bill} = \text{Sale Amount} - \text{Discount Amount}$
9. Display the customer details, product details, quantity, sale amount, discount, and final bill.
10. End.

Flowchart – Shop Management System (Feature Set III)



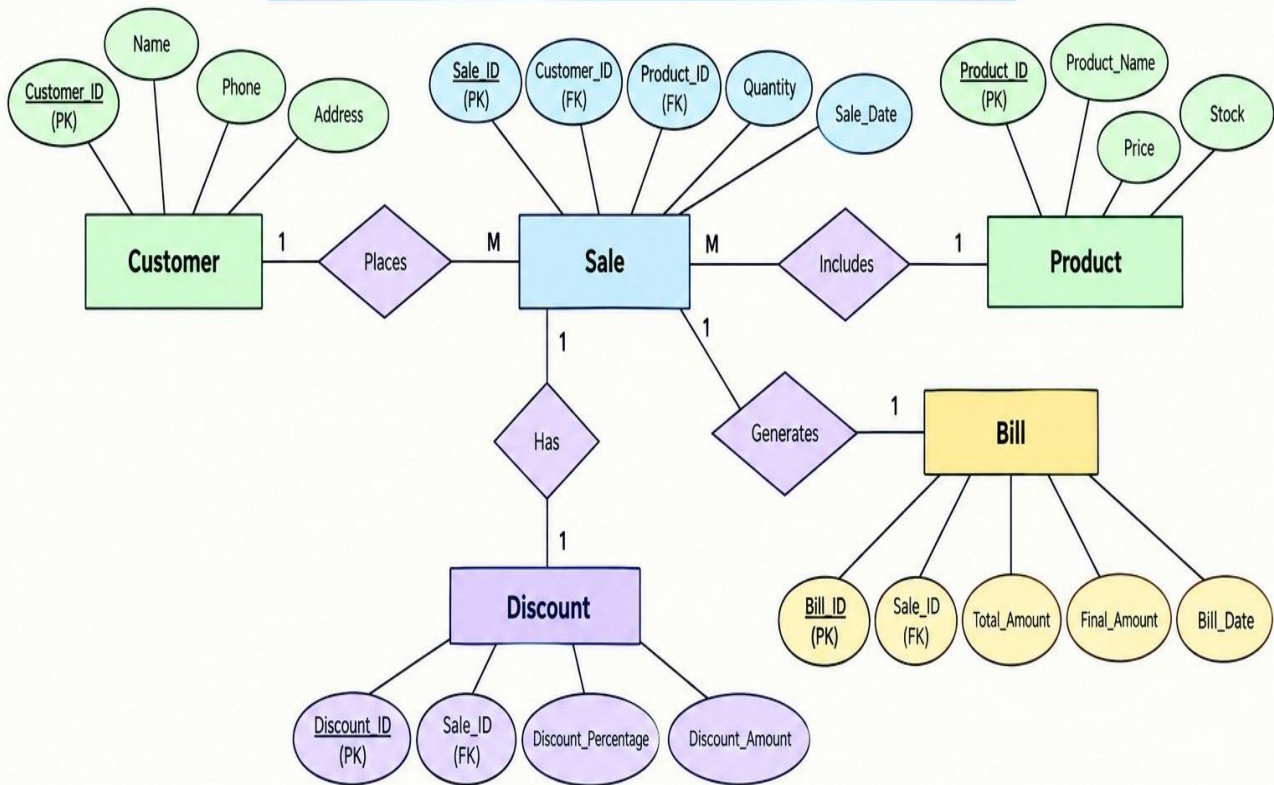
Flowchart Symbols



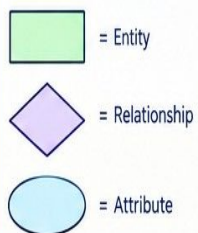
Feature Set III

1. Customer details
2. Product details
3. Sale
4. Discount
5. Final bill

ER Diagram – Shop Management System (Feature Set III)



Legend



PK = Primary Key
(underlined)
FK = Foreign Key
(marked as FK)
1:M = One to Many
1:1 = One to One

Relationships & Cardinality

1. Customer → Places → Sale (1 : M) (One customer can place many sales, but each sale is made by one customer)
2. Product → Includes → Sale (1 : M) (One product can be in many sales, but each sale contains one product)
3. Sale → Has → Discount (1 : 1) (Each sale has one discount and each discount is for one sale)
4. Sale → Generates → Bill (1 : 1) (Each sale generates one bill and each bill is for one sale)